DEIJANY RODRIGUEZ LINARES

Ph.D. in Electrical Engineering — DSP & Communication Systems

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PROFILE

DSP/ML engineer with a Ph.D. in Electrical Engineering, specialized in low-complexity, hardware-aware signal processing for wireless communication systems (ELLIIT project: Baseband Processing for Beyond 5G Wireless). Designs and validates algorithms for synchronization, linearization, and equalization under real hardware constraints, balancing accuracy, complexity, and deployability, including RTL design and FPGA synthesis experience.

EXPERIENCE

Doctoral Researcher – DSP & Communication Systems

Linköping University

2021 – 2026 Linköping, Sweden

- Developed efficient sampling-frequency offset (SFO) estimation and compensation algorithms for wideband systems, reusing the same Farrow structure across both stages to substantially reduce computational complexity versus conventional approaches
- Developed a unified analytical nullspace-based least-squares design for the Farrow structure, enabling mixed linear-phase and general FIR subfilters required for delay constrained applications
- Derived a cascaded-accumulator based method for efficient real-time computation of time-index powered weighted sums common to multiple SFO estimation approaches (Newton, GD, iterative LS, LMS), cutting multiplications from scaling with sequence length to a small constant number; validated via RTL design and FPGA synthesis (Verilog, Yosys, Project Trellis) targeting a Lattice ECP5
- Designed a family of low-complexity, learning-inspired linearizers for ADC nonlinearities (memoryless and frequency-dependent), trained via convex optimization, matching neural-network-based accuracy at a fraction of the computational cost
- Designed a memoryless ADC linearizer using 1-bit quantization, requiring only $B+1$ lookup-table entries for a B-bit output resolution (versus 2^B for a direct table-based approach), reducing computation to a single lookup and multiplication per sample
- Extended linear-phase FIR filter design for DAC frequency-response equalization beyond prior work, limited to the first Nyquist bands, to a minimax-optimal design covering the first 6 Nyquist bands, enabling the order-estimation method developed next
- Derived a closed-form symbolic-regression-based estimate for the FIR filter order in DAC equalization, meeting target specifications across multiple Nyquist bands, eliminating manual trial-and-error tuning
- Teaching assistant for three DSP/communications courses: Signal Processing for Communications, Analog Filters, and Digital Filters

Teaching and Research Assistant

University of Havana (InSTEC)

2019 – 2020 Havana, Cuba

- Combined deep learning with Monte Carlo radiation transport to improve prediction accuracy for low-probability interaction events
- Served as teaching assistant for Numerical Mathematics II and Fundamentals of Medical Physics

SKILLS

Signal Processing · Optimization · MIMO · Estimation & Synchronization · OFDM · Linearization · RTL / FPGA Synthesis · Hardware-Aware DSP · FIR Filter Design · Adaptive Filtering · Channel Estimation

TECHNICAL SKILLS

MATLAB / Python
C++ / CUDA / Git / SQL
Linux / Bash
Verilog / FPGA toolflow

TOOLS / FRAMEWORKS

PyTorch, TensorFlow (incl. Keras), NumPy, SciPy, scikit-learn, JAX, SymPy, MLX
Yosys, nextpnr, Project Trellis (ECP5), Vivado
GNU Radio, Geant4, EGSnrc
Docker, Docker Compose, CI/CD (GitLab CI)

EDUCATION

Ph.D. in EE (Comm. Systems)

Linköping University

2021–2026 Sweden

- Thesis: Contributions to Low-Complexity Linearization, Equalization, and Synchronization
- Relevant coursework: Modern Radar Systems, Detection and Estimation of Signals, System Identification, Multirate Digital Signal Processing, Signal Processing for Communications, Matrix Analysis, Nonlinear Optimization, Reinforcement Learning

M.Sc. in Nuclear Engineering (summa cum laude)

University of Havana (InSTEC)

2016–2018 Cuba

- Thesis: Deep learning for image recognition and breast tumor classification from X-ray sources

Diploma in Medical Physics

University of Havana (InSTEC)

2015–2016 Cuba

- Thesis: Monte Carlo–based pretreatment verification for IMRT

Medical Physicist — QA and Modeling

CECMED

2015 – 2018 Havana, Cuba

- Developed a CNN for X-ray tumor detection and severity classification, using transfer learning and data augmentation (rotation, color masking, horizontal flipping) to address limited medical imaging data, achieving 91.43% and 86.67% recall, respectively
- Applied classical image processing techniques, including PCA-based analysis, to support tumor imaging assessment and dose planning decisions
- Performed Monte Carlo simulation of radiation transport (GATE, Geant4, EGSnrc) for phantom-based dose verification, supporting treatment planning system validation
- Led at least 15 radiotherapy quality assurance audits as CECMED's regulatory representative across national radiotherapy services, collaborating with hospital-based medical physicists, covering linac, cobalt-60, and superficial X-ray equipment
- Co-mentored two undergraduate thesis projects on Monte Carlo simulation of linac and X-ray machines

PUBLICATIONS

Author of 7 peer-reviewed publications (4 journal articles, 3 conference papers), plus 2 papers currently under review at IEEE Signal Processing Letters and Signal Processing Open.

Full list at `deijany.com/publications`.

SELECTED PROJECTS

`CNN-linearizer:` CNN-based ADC linearizer (parallel ReLU/abs-filter branches without crossconnections), the precursor to the low-complexity analytical solution (IEEE Access 2025)

`RL-connecting-four:` Trained three PPO variants (SB3 masked, SB3 vanilla, from-scratch PyTorch) via self-play on Connect Four (PettingZoo)

See more projects at `github.com/deijany` (see pinned repositories)

CERTIFICATIONS

Deep Learning Specialization (DeepLearning.AI)
TensorFlow Developer Professional Certificate (DeepLearning.AI)

AWARDS

Excellent Oral Presentation Award, IEEE ICCT 2024
ICTP Travel Award and Full Scholarship (2017, 2018, 2020)
National Scholarship of the Ministry of Education, Cuba (2010-2015)

B.Sc. in Energy & Nuclear Technology

University of Havana (InSTEC)

2010–2015 Cuba

- Thesis: Optimization of a quality control procedure for IMRT

TECHNICAL TRAINING
(EXAM- OR PROJECT-BASED)

ML for Dark Matter Search

ICTP

2019 Trieste, Italy

- Project-based ML methods for weak-signal detection in large-scale, low-SNR data

Parallel and Distributed Computing

ICIMAF

2018 Havana, Cuba

- Exam-based training in parallel programming and distributed computing

Scientific Software Development

Sharif University of Technology & ICTP

2018 Tehran, Iran

- Project-based training in collaborative open-source scientific software development

Monte Carlo Simulation

ICTP & IAEA

2017 Trieste, Italy

- Advanced training in Monte Carlo simulation and modeling techniques (based on EGSnrc)

MEMBERSHIP

Junior Associate, ICTP (2019–2024)

LANGUAGES

Spanish (Native)
English (Fluent)

REFEREES

Prof. Håkan Johansson (Ph.D. supervisor)
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Further references available upon request